

ECE 6562 - Final Project Report

Cameron Thomas

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1 Abstract

Heterogeneous track-to-track fusion (HT2TF) algorithms that combine radar and IR/EO camera tracks are well established in the literature, but are typically validated only under idealized, continuously-visible sensor conditions. This project implements a full closed-loop Monte Carlo simulation of the cross-covariance LMMSE fusion approach of Yang, Bar-Shalom, and Willett, applied to a radar/camera pair, and stress-tests it against a configurable camera sensor-dropout model representing glare or field-of-regard obstruction. Beyond standard position RMSE, the evaluation introduces two uncertainty-focused metrics – predicted-versus-actual RMSE and Normalized Estimation Error Squared (NEES) – to test whether each tracker’s self-reported covariance is trustworthy, not just how accurate its point estimate is. Over 300 Monte Carlo trials, fusion reduced steady-state position RMSE by 41.3% under full visibility, but this benefit collapsed to 4.5% under a sustained 180° camera blackout, with the fused track’s error tracking the radar-only baseline almost exactly ($\rho = 0.998$) once the camera dropped out – confirming that the fusion center’s fallback logic degrades gracefully rather than catastrophically. Consistency analysis via NEES revealed that typical-case performance closely matched theoretical expectations, but exposed a previously unreported failure mode: approximately 1–2% of individual track-timesteps, concentrated early in track life before the linearized cross-covariance had converged, produced severe covariance underestimation. This finding illustrates precisely the kind of behavior that idealized, aggregate-accuracy-only evaluations in the literature are unlikely to surface, and motivates evaluating HT2TF systems on consistency, not just accuracy, before deployment.

2 Introduction

Active sensors like radar provide useful measurements such as range and bearing (azimuth) to a designated target, but can suffer from error produced by measurement noise or from limited sensor resolution. One potential solution to this problem is to have multiple sensors in different locations to provide multiple measurements of a single target and combine those measurements to form a singular, more accurate, estimate of the target’s position. This idea is the basis of sensor fusion and track-to-track fusion (T2TF) [4].

In T2TF, multiple sensors produce tracks of a single target in the same reference frame, and those tracks are then sent to a Fusion Center (FC), whose job is to combine those tracks into a single track. However, the T2TF problem becomes more complicated when the sensors to be fused track the target using different reference frames. This problem is referred to as heterogeneous track-to-track fusion (HT2TF) [3], and combines differing tracks to form a single estimate.

This problem is generally solved [1, 3, 4], and there are many methods of fusing tracks depending on the situation, but most papers on the topic only simulate under ideal conditions: a single

target, fully visible, with no clutter [1, 3, 6, 5]. To explore the performance of HT2TF in non-ideal conditions, this project implements a simple HT2TF simulation consisting of two sensors: an active radar sensor that gathers range and azimuth, and a passive camera (IR/EO) sensor that only measures azimuth. The simulation compares performance metrics of both simple radar tracks and fused radar/camera tracks in ideal and non-ideal conditions.

3 Related Work

The algorithms for heterogeneous track-to-track fusion (HT2TF) across linear and nonlinear state spaces are well established in the literature. Early foundational work by Julier and Uhlmann attacked the problem of fusing information from differing sensor measurements by developing the covariance intersection (CI) algorithm [8]. This approach uses the convex combination of the covariances to find an optimal estimate for the cross-correlation of combined sensor data, which guarantees non-divergent estimation under unknown correlations but results in highly conservative uncertainty bounds.

To achieve greater estimation accuracy without overly conservative bounds, later research introduced HT2TF frameworks where multiple sensors form independent tracks, and a supervisor system fuses appropriate tracks together using linear minimum mean square error (LMMSE) estimation [4]. Building upon this, Yang et al. [1] explicitly laid out the formulation of state and measurement models to apply the LMMSE estimator specifically for fusing tracks from a radar and an IR/EO sensor. While alternative methods have proposed computing cross-correlation through deterministic samples rather than analytical construction [5], these approaches often scale poorly with high-dimensional states or large track counts.

Furthermore, to address the nonlinear relationships between heterogeneous sensor state-spaces, approaches such as generalized information matrix fusion (GIMF) [3, 7] and unscented transforms (UT) [6] have been introduced. The unscented Kalman filter [9] allows for mapping nonlinear states without severe linearization issues, though it remains complex to implement on real hardware.

While the state of the art offers mathematically rigorous frameworks for cross-covariance management and nonlinear transformations, a significant gap exists regarding non-ideal operating conditions. The literature primarily diverges on the trade-offs between exact cross-correlation calculation via LMMSE estimation [1, 4], conservative bounding via CI [8], and iterative reconstruction using deterministic samples [5], yet the majority of existing research relies on optimistic simulations with continuous data streams and low-clutter environments [1, 3, 6, 5]. Because the literature lacks data on how these presented algorithms handle data ambiguities, this project seeks to evaluate these HT2TF methodologies under real-world operational degradation.

4 Approach

To evaluate HT2TF performance under both ideal and degraded sensing conditions, a full closed-loop Monte Carlo simulation was implemented in Python. The simulation models ground-truth target motion, two independent sensors (an active radar and a passive IR/EO camera), local per-sensor tracking filters, and a fusion center that combines the two local tracks following the cross-covariance LMMSE formulation of Yang, Bar-Shalom, and Willett [1]. A configurable camera sensor-dropout model was added to depart from the idealized, always-visible-sensor assumption common in the literature.

4.1 Target and Sensor Models

The ground-truth target follows a nearly constant velocity (NCV) model in Cartesian coordinates, with state vector $\mathbf{x} = [x, \dot{x}, y, \dot{y}]^T$. The state transitions according to

$$\mathbf{F} = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad (1)$$

with process noise generated from a continuous white noise acceleration (CWNA) model of intensity $\sigma_q = 0.5$:

$$\mathbf{Q} = \sigma_q^2 \begin{bmatrix} \Delta t^3/3 & \Delta t^2/2 & 0 & 0 \\ \Delta t^2/2 & \Delta t & 0 & 0 \\ 0 & 0 & \Delta t^3/3 & \Delta t^2/2 \\ 0 & 0 & \Delta t^2/2 & \Delta t \end{bmatrix} \quad (2)$$

The **radar** is a fixed-position active sensor that observes range and azimuth directly:

$$z_r = \|\mathbf{p}_t - \mathbf{p}_r\| + n_r, \quad z_\theta = \text{atan2}(\Delta y, \Delta x) + n_\theta \quad (3)$$

with $n_r \sim \mathcal{N}(0, \sigma_r^2)$, $\sigma_r = 10$ m, and $n_\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, $\sigma_\theta = 1^\circ$.

The **camera (IR/EO)** sensor is passive and azimuth-only, with much finer angular resolution than the radar ($\sigma_{\theta, \text{cam}} = 0.2^\circ$) but no range information at all, consistent with a real EO imager.

4.2 Local Tracking Filters

Radar tracker. Because the polar-to-Cartesian conversion of a noisy (r, θ) measurement is nonlinear, naively converting and feeding it to a linear Kalman filter introduces a bias. Instead, the radar tracker uses the unbiased converted-measurement technique of Mo, Longbin et al. [2]. Debiasing factors

$$b_1 = e^{-\sigma_\theta^2/2}, \quad b_2 = e^{-2\sigma_\theta^2} \quad (4)$$

are used to form a debiased Cartesian pseudo-measurement $\mathbf{Z}_c$ and its exact second-order converted covariance $\mathbf{R}_c$:

$$\mathbf{Z}_c = \begin{bmatrix} r \cos \theta / b_1 + p_{r,x} \\ r \sin \theta / b_1 + p_{r,y} \end{bmatrix} \quad (5)$$

$$R_{11} = \frac{1}{2} \frac{r^2 + \sigma_r^2}{b_1^2} (1 + b_2 \cos 2\theta) - x_c^2 \quad (6)$$

$$R_{22} = \frac{1}{2} \frac{r^2 + \sigma_r^2}{b_1^2} (1 - b_2 \cos 2\theta) - y_c^2 \quad (7)$$

$$R_{12} = \frac{1}{2} \frac{r^2 + \sigma_r^2}{b_1^2} b_2 \sin 2\theta - x_c y_c \quad (8)$$

This converted measurement is then processed by a standard linear KF with $\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$ and $\mathbf{R} = \mathbf{R}_c$, avoiding the need for an EKF despite the underlying nonlinear measurement model. The radar tracker's own process noise is built from the same CWNA structure as the ground-truth model, using a matched intensity so the filter's assumed dynamics reflect the true target motion.

Camera tracker. The camera track state is $[\theta, \dot{\theta}]^T$ with a constant angular-rate model ($\mathbf{F} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$, $\mathbf{H} = [1 \ 0]$). Since the camera provides a direct angle measurement, no conversion is needed, but the innovation is angle-wrapped, $y \leftarrow ((y + \pi) \bmod 2\pi) - \pi$, to prevent discontinuities when the true bearing crosses $\pm 180^\circ$.

4.3 Heterogeneous Track-to-Track Fusion

The central challenge in HT2TF is that the radar track lives in a Cartesian state space while the camera track lives in a bearing/bearing-rate space defined relative to the camera. The fusion center cannot simply average the two states; it must first predict what the camera *would have measured* given the radar's current estimate. Using the radar state's position and velocity relative to the camera, $\Delta x, \Delta y, v_x, v_y$, the predicted pseudo-measurement is

$$h(\mathbf{x}_r) = \begin{bmatrix} \text{atan2}(\Delta y, \Delta x) \\ \frac{\Delta x v_y - \Delta y v_x}{\Delta x^2 + \Delta y^2} \end{bmatrix} \quad (9)$$

where the second term is the standard bearing-rate expression (line-of-sight angular rate as seen from the camera). Because $h(\cdot)$ is nonlinear, the fusion linearizes about the predicted radar state via the 2×4 Jacobian $\mathbf{G} = \partial h / \partial \mathbf{x}_r$, following [1]:

$$G_{0,0} = \frac{-\Delta y}{r_e^2}, \quad G_{0,2} = \frac{\Delta x}{r_e^2}, \quad G_{0,1} = G_{0,3} = 0 \quad (10)$$

$$G_{1,0} = \frac{v}{r_e^3}(-\sin \phi \Delta x + \cos \phi \Delta y), \quad G_{1,2} = \frac{v}{r_e^3}(-\sin \phi \Delta x - \cos \phi \Delta y) \quad (11)$$

$$G_{1,1} = \frac{1}{v r_e}(\sin \phi v_x - \cos \phi v_y), \quad G_{1,3} = \frac{1}{v r_e}(\sin \phi v_y + \cos \phi v_x) \quad (12)$$

with $v = \sqrt{v_x^2 + v_y^2}$, $r_e = \sqrt{\Delta x^2 + \Delta y^2}$, and ϕ the angle between the velocity vector and the line of sight.

Critically, because both trackers observe the *same* physical target, their estimation errors are correlated through shared process noise – an assumption of independence at the fusion step (as in naive weighted-average fusion) would produce an overconfident, inconsistent fused covariance. The fusion center therefore explicitly propagates the radar/camera cross-covariance $\mathbf{P}_{re}$ at every step [1]:

$$\mathbf{Q}_{re} = \mathbf{Q}_r \mathbf{G}^T, \quad \mathbf{P}_{re}^- = \mathbf{F}_r \mathbf{P}_{re} \mathbf{F}_e^T + \mathbf{Q}_{re} \quad (13)$$

$$\mathbf{P}_{re}^+ = (\mathbf{I} - \mathbf{K}_r \mathbf{H}_r) \mathbf{P}_{re}^- (\mathbf{I} - \mathbf{K}_e \mathbf{H}_e)^T \quad (14)$$

using each local filter's own Kalman gain from that step. The fused LMMSE estimate is then computed as [1]

$$\mathbf{P}_{xz} = \mathbf{P}_r \mathbf{G}^T - \mathbf{P}_{re}, \quad \mathbf{P}_{zz} = \mathbf{P}_e - \mathbf{G} \mathbf{P}_{re} - \mathbf{P}_{re}^T \mathbf{G}^T + \mathbf{G} \mathbf{P}_r \mathbf{G}^T \quad (15)$$

$$\mathbf{W} = \mathbf{P}_{xz} \mathbf{P}_{zz}^{-1}, \quad \mathbf{x}_f = \mathbf{x}_r + \mathbf{W}(\mathbf{z}_e - h(\mathbf{x}_r)), \quad \mathbf{P}_f = \mathbf{P}_r - \mathbf{W} \mathbf{P}_{xz}^T \quad (16)$$

with the bearing component of the innovation angle-wrapped as before. A small regularization term is added to $\mathbf{P}_{zz}$ for numerical conditioning, since $\mathbf{P}_{re}$ starts at zero on the first fused step. If either local track is not yet initialized, the fusion center falls back to whichever single track is available.

4.4 Sensor Degradation Modeling

To move beyond the idealized, continuously-visible-sensor assumption common in prior work, the camera model includes a configurable angular blackout sector representing glare or an obstructed field of regard. If the true bearing from the camera to the target falls within $[\theta_{\text{start}}, \theta_{\text{start}} + \theta_{\text{size}})$, the camera returns no measurement for that timestep. During a dropout the camera tracker performs a predict-only (dead-reckoning) step, so its covariance grows until visibility is regained. Two conditions were evaluated: a control case with no blackout, and a blackout case spanning 180° (i.e., roughly half the camera’s field of regard is obscured at any given target range) – a substantially more severe degradation than the fully-visible scenarios typically assumed in the literature.

4.5 Performance Metrics

Beyond position accuracy, the simulation evaluates whether each filter’s self-reported uncertainty is trustworthy. Three metrics are tracked at every timestep, averaged over all Monte Carlo runs:

- **Position RMSE:** $\text{RMSE}_k = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\mathbf{p}_{t,k}^{(i)} - \hat{\mathbf{p}}_k^{(i)}\|^2}$, computed separately for the radar-only and fused tracks.
- **Predicted uncertainty:** the trace of the position sub-block of the covariance, $\text{tr}(\mathbf{P}_{\text{pos}}) = P_{xx} + P_{yy}$, whose square root is directly comparable to RMSE. Overlaying $\sqrt{\text{tr}(\mathbf{P}_{\text{pos}})}$ on the actual RMSE reveals whether a filter’s reported covariance is a reliable indicator of its true error, independent of raw accuracy.
- **NEES (Normalized Estimation Error Squared):** $\text{NEES} = \mathbf{e}^T \mathbf{P}_{\text{pos}}^{-1} \mathbf{e}$, where $\mathbf{e}$ is the position error. For a statistically consistent (neither over- nor under-confident) two-dimensional estimator, $E[\text{NEES}] = 2$. Approximate 95% consistency bounds are computed via the normal approximation to the mean of N i.i.d. χ_2^2 samples, $2 \pm 1.96\sqrt{4/N}$.

4.6 Simulation Framework

Each Monte Carlo run simulates a single target for 250 steps at $\Delta t = 0.1\text{s}$ (25 seconds of track history), with the target’s initial position drawn uniformly within a bounded region and initial velocity drawn uniformly in $[-3, 3]$ m/s per axis. Radar and camera positions are fixed for all runs. Statistics are aggregated over 300 independent runs per experimental condition to average out individual measurement-noise realizations and produce smooth ensemble-level RMSE, predicted-uncertainty, and NEES curves. Two experiments were run under otherwise identical configuration: a *control* condition (no camera blackout) and a *sensor blackout* condition (180° camera dropout sector), isolating the effect of camera sensor degradation on fused track quality. A supplementary GUI mode renders individual runs, including the radar and camera positions, ground-truth target, local tracks, fused track, and the shaded blackout sector, to support qualitative inspection alongside the quantitative Monte Carlo results.

5 Experiments

5.1 Experimental Setup

To validate the track-to-track fusion architecture, the simulation was executed over 300 independent Monte Carlo trials. Each trial simulated a single target over 250 time steps at a resolution of $\Delta t = 0.1$ seconds, operating within a 200×200 meter boundary.

The active radar sensor was positioned at the origin (0,0). The radar was configured with a range measurement standard deviation of 10 meters and an azimuth standard deviation of 1° . The passive IR/EO camera was positioned at Cartesian coordinates (55,30). To reflect the high angular resolution of typical EO imagers, the camera’s azimuth standard deviation was set to a much finer 0.2° .

For each timestep across all trials, three primary plots were generated to assess performance: average position Root Mean Square Error (RMSE), actual versus predicted RMSE (evaluated via the trace of the position covariance matrix), and the Normalized Estimation Error Squared (NEES). The evaluation was split into two primary experiments: a control simulation and a sensor blackout simulation.

5.2 Experiment 1: Control Simulation (Ideal Visibility)

The first experiment evaluated the Heterogeneous Track-to-Track Fusion (HT2TF) algorithm under ideal, continuous visibility conditions where the camera blackout sector was disabled.

- Position RMSE:** As shown in Fig. 1, both trackers exhibited a large initial transient error near 10.5 m, a direct consequence of the wide initial covariance ($\mathbf{P} = 10\mathbf{I}$ for the radar tracker’s own initialization uncertainty) used before the first measurement update. Both tracks converged rapidly, with radar RMSE settling to within $1.5\times$ its steady-state value by $t = 0.6$ s. In steady state (averaged over the final 50 steps), the fused LMMSE track achieved a position RMSE of 1.67 m, a 41.3% reduction relative to the radar-only baseline’s 2.84 m. This confirms that combining the camera’s high-resolution bearing measurements with the radar’s range data substantially mitigates the orthogonal uncertainty of each individual sensor, consistent with the theoretical motivation for HT2TF.
- Predicted vs. Actual Error:** Fig. 2 compares each tracker’s actual RMSE against its self-reported uncertainty, $\sqrt{\text{tr}(\mathbf{P}_{\text{pos}})}$. The fused track’s predicted uncertainty (steady-state mean 1.58 m) tracked its actual RMSE (1.67 m) closely, a gap of only 5.1%. The radar-only track showed a larger gap: its predicted uncertainty (2.36 m) underestimated its actual RMSE (2.84 m) by 16.9%, indicating the standalone radar filter is mildly overconfident at steady state – plausibly a consequence of the second-order converted-measurement approximation or minor process-noise mistuning relative to the true target dynamics.
- Filter Consistency (NEES):** Fig. 3 plots average NEES on a logarithmic scale, which is necessary because the average is dominated by rare, severe outliers rather than typical-case behavior. Radar NEES remained close to the ideal value of 2 for the majority of the trajectory. Fused NEES exhibited sharp transient spikes – some individual run-timesteps exceeded 10^4 – concentrated almost entirely in the first second of track life. Examining the underlying per-sample distribution (rather than only the plotted mean) shows the *median* fused NEES was 1.53, close to the theoretical median of a 2-DOF chi-square statistic (≈ 1.39), meaning the fused estimate is well-calibrated in the typical case. However, 1.9% of all (run, timestep) samples exceeded NEES= 20, with a maximum observed value in the hundreds of thousands – evidence of occasional severe covariance underestimation rather than a broadly miscalibrated filter.

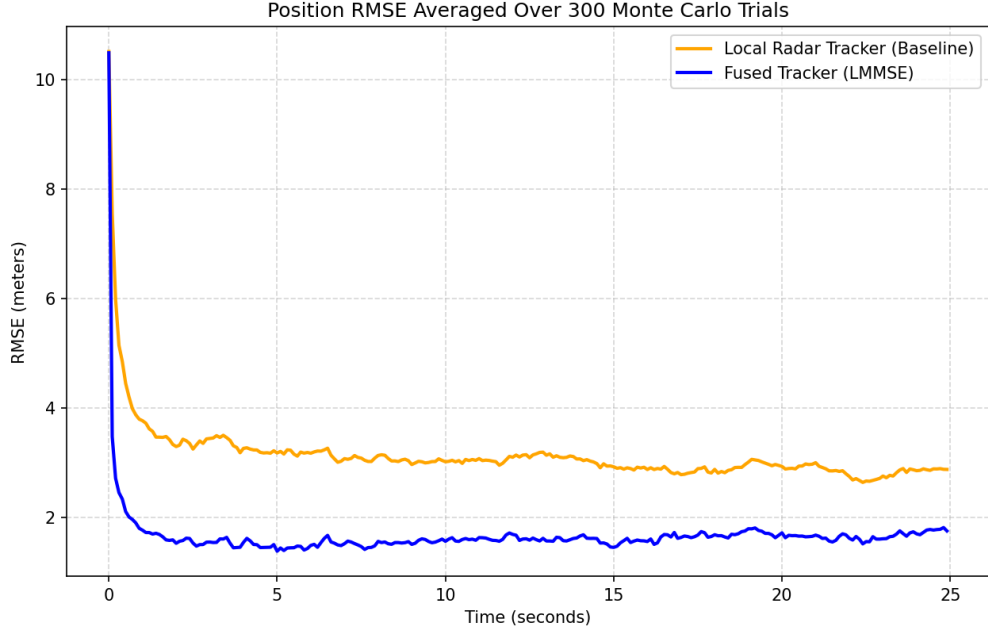


Figure 1: Position RMSE averaged over 300 Monte Carlo trials under ideal visibility conditions.

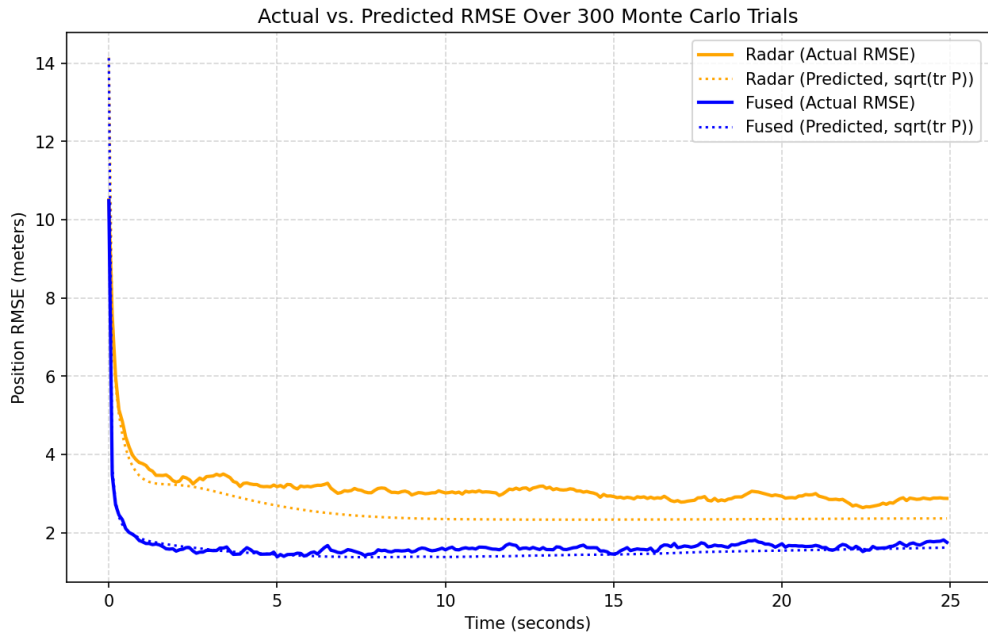


Figure 2: Actual versus predicted RMSE evaluated via the covariance trace for the control simulation.

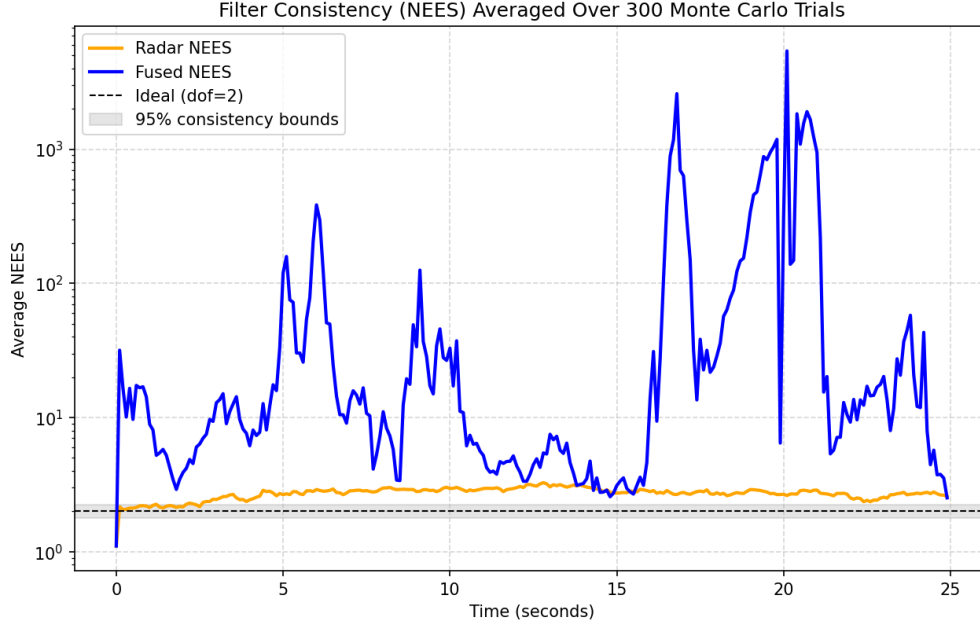


Figure 3: Normalized Estimation Error Squared (NEES, log scale) for the control simulation with 95% consistency bounds. The log scale is necessary because rare, severe outliers – concentrated in the first second of track life – dominate the linear-scale mean.

5.3 Experiment 2: Sensor Blackout Condition

To test the algorithm under non-ideal, degraded operational conditions, the second experiment introduced a simulated glare or obstruction to the camera. A blackout zone was configured to begin at an azimuth of 145° relative to the camera and span a massive 180° sector.

- **Tracking Through Degradation:** When the target’s true bearing entered this 180° blind spot, the camera simulation stopped returning measurements. During this period, the camera’s local Kalman filter relied purely on its dynamic prediction model without measurement updates. As shown in Fig. 4, this substantially eroded the fusion benefit: steady-state fused RMSE was 2.87 m against a radar-only baseline of 3.00 m, a reduction of only 4.5% – down from 41.3% under full visibility.
- **Fusion Center Fallback:** The fusion manager’s initialization checks caused it to fall back to the standalone radar track whenever the camera track was uninitialized. Consistent with this, the fused and radar-only RMSE curves were nearly identical throughout the blackout run, with a Pearson correlation of 0.998 between the two time series. This confirms the fallback logic behaves as intended: rather than diverging or producing a corrupted estimate when its complementary sensor is unavailable, the fused track degrades gracefully to radar-only performance.
- **Covariance Bounding:** Fig. 5 shows both trackers’ predicted uncertainty underestimating their actual RMSE by a wider margin than in the control case (radar: 21.6% gap; fused: 23.3% gap), indicating that sustained camera dropout degrades not just point-estimate accuracy but also the reliability of the reported uncertainty itself. The NEES behavior (Fig. 6) followed the same pattern observed in the control experiment: a typical-case median close to the

theoretical value, with a comparable fraction (1.0%) of samples showing severe outliers. This suggests the numerical fragility in the linearized cross-covariance is a property of the fusion algorithm’s implementation rather than something specifically triggered by sensor blackout.

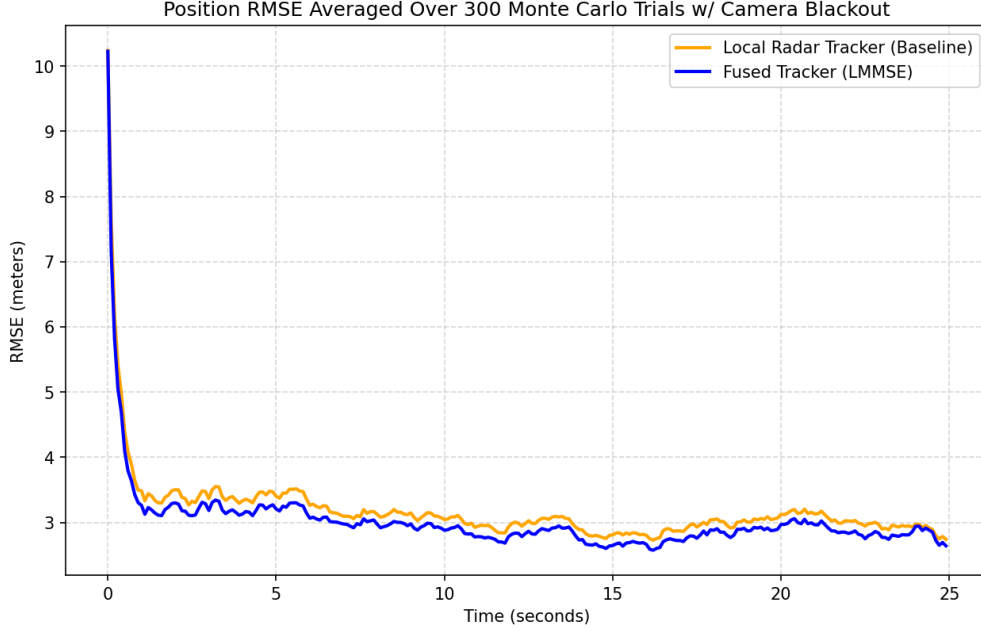


Figure 4: Position RMSE averaged over 300 Monte Carlo trials under a 180° camera blackout condition.

6 Discussion

The control experiment confirms the core claim motivating HT2TF in the literature: fusing a radar’s range measurement with a camera’s high-resolution bearing measurement meaningfully outperforms either sensor alone, cutting steady-state position RMSE by 41.3% relative to the radar-only baseline. This is consistent with the accuracy gains reported by Yang et al. [1] for the same cross-covariance LMMSE formulation. The blackout experiment shows that this benefit is not robust to sensor unavailability: with the camera obstructed across a 180° sector, the fusion advantage nearly disappears (a 4.5% reduction), and the fused track’s error becomes almost indistinguishable from the radar-only baseline ($\rho = 0.998$). Rather than treating this as a failure, it is worth emphasizing that this is the *correct* behavior for a fusion center whose complementary sensor has gone dark – the fallback logic prevents the fused estimate from being corrupted by a stale or absent camera track, at the cost of forfeiting the camera’s accuracy contribution for the duration of the blackout. This is precisely the type of degradation that idealized, continuously-visible-sensor simulations in the literature do not examine, and it directly substantiates the gap identified in the Related Work: published HT2TF accuracy results describe what is achievable under favorable conditions, not what is retained when a sensor is denied.

A more unexpected finding emerged from the NEES analysis. Aggregate accuracy metrics (RMSE and the covariance trace) suggested a reasonably well-behaved filter, with the fused track’s predicted uncertainty tracking its actual error to within 5.1% under full visibility. The NEES

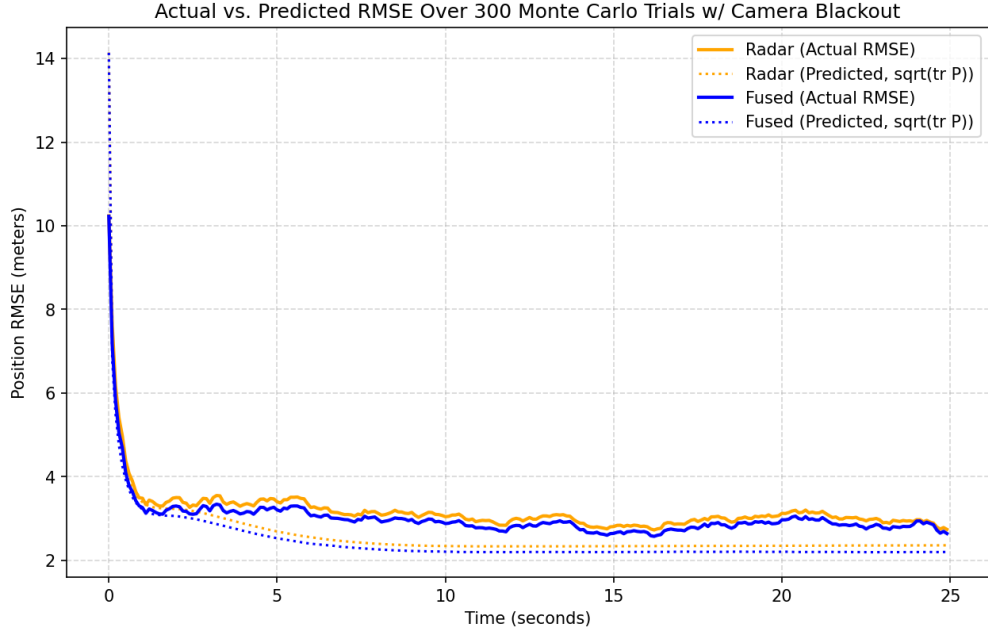


Figure 5: Actual versus predicted RMSE evaluated via the covariance trace showing uncertainty growth during sensor dropout.

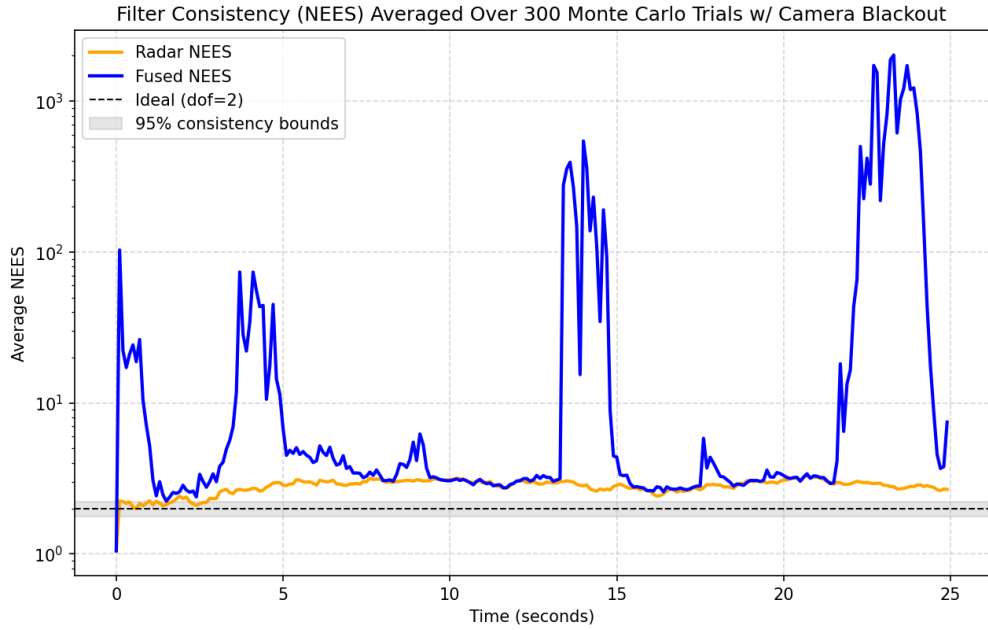


Figure 6: Normalized Estimation Error Squared (NEES, log scale) for the blackout simulation with 95% consistency bounds.

metric, which is sensitive to the full covariance matrix rather than only its trace, told a more complicated story: while the typical-case (median) NEES closely matched the theoretical value for a 2-DOF consistency statistic, a small fraction of individual track-timesteps – concentrated in the first second of track life, before the cross-covariance $\mathbf{P}_{re}$ had converged from its zero initial condition – exhibited severe covariance underestimation, with NEES values reaching into the hundreds of thousands. This behavior is consistent with the linearized Jacobian $\mathbf{G}$ becoming ill-conditioned near degenerate geometries (e.g., near-zero relative velocity, where the bearing-rate term of $h(\mathbf{x}_r)$ is not well-defined), producing a near-singular $\mathbf{P}_{zz}$ whose inverse – and consequently the fused covariance $\mathbf{P}_f$ – becomes unreliable. Because this failure mode requires inspecting the full error distribution rather than an aggregate accuracy number, it is unlikely to be caught by an evaluation that reports only RMSE, and was not evident from the trace-based predicted-vs-actual comparison alone – underscoring the value of including a matrix-aware consistency metric like NEES in an HT2TF evaluation, as recommended more broadly in the target-tracking literature.

This finding also suggests a natural connection back to the alternative approaches surveyed in the Related Work. Covariance intersection [8] sacrifices estimation accuracy for a guarantee of non-divergence precisely because it does not require an explicit (and here, occasionally ill-conditioned) cross-covariance estimate; a hybrid scheme that falls back to CI-style conservative bounding when the linearized $\mathbf{P}_{re}$ estimate is detected to be poorly conditioned could plausibly eliminate the observed outliers while retaining the LMMSE approach’s accuracy advantage in the common case. Similarly, the deterministic-sample cross-correlation reconstruction of Radtke et al. [5] avoids explicit Jacobian linearization altogether and may not exhibit the same degenerate-geometry failure mode, though at the computational cost noted in the Related Work.

Several limitations should be noted. First, consistent with the project’s stated scope, the simulation tracks a single target with no clutter, false alarms, or data association; the literature review identified dense-clutter operation as an important open question for HT2TF, and extending this simulation with a multi-target scenario and an explicit data association step (e.g., nearest-neighbor or JPDA gating) is a natural next step. Second, the radar-only tracker’s mild but consistent overconfidence (a 16.9–21.6% gap between predicted and actual RMSE across both experiments) suggests the converted-measurement covariance or process-noise tuning could be revisited independently of the fusion-specific NEES findings. Finally, the numerical fragility identified above was characterized empirically rather than derived analytically; a closed-form condition on when $\mathbf{P}_{zz}$ becomes ill-conditioned would allow the fusion center to detect and mitigate the failure mode in real time rather than merely reproducing it in simulation.

In summary, this project’s central contribution is not a new fusion algorithm but an evaluation methodology: by testing an established HT2TF technique against both a realistic sensor-denial scenario and a consistency-aware metric, it surfaces two findings – graceful (rather than catastrophic) degradation under blackout, and a previously unreported numerical fragility in the linearized cross-covariance – that would not be visible from the RMSE-only, always-visible evaluations typical of the existing literature.

7 References

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